

## Git HOL1

```
MINGW64:/c/Users/kpre0

kpre0@DESKTOP-L14143A MINGW64 ~
$ git --version
git version 2.47.0.windows.1
```

```
kpre0@DESKTOP-L14143A MINGW64 ~
$ git config --global user.name="ThePreethi"

kpre0@DESKTOP-L14143A MINGW64 ~
$ git config --global user.email="preethi4u97@gmail.com"

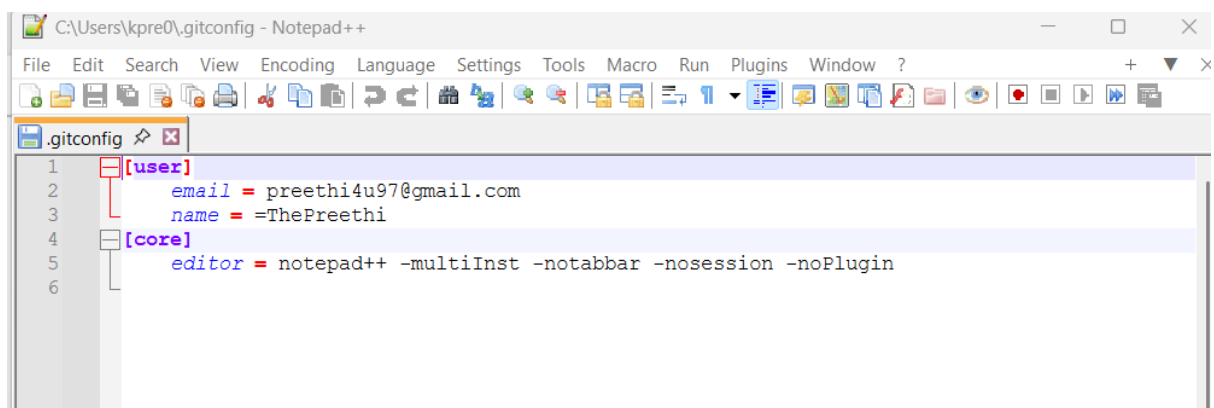
kpre0@DESKTOP-L14143A MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.email=preethi4u97@gmail.com
user.name=ThePreethi

kpre0@DESKTOP-L14143A MINGW64 ~
$ notepad++

kpre0@DESKTOP-L14143A MINGW64 ~
$

kpre0@DESKTOP-L14143A MINGW64 ~
$ git config --global core.editor "notepad++ -multiInst -notabbar -nosession -no
Plugin"

kpre0@DESKTOP-L14143A MINGW64 ~
$ git config --global -e
hint: Waiting for your editor to close the file... |
```



```
C:\Users\kpre0\.gitconfig - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

1 [user]
2   email = preethi4u97@gmail.com
3   name = ThePreethi
4 [core]
5   editor = notepad++ -multiInst -notabbar -nosession -noPlugin
6
```

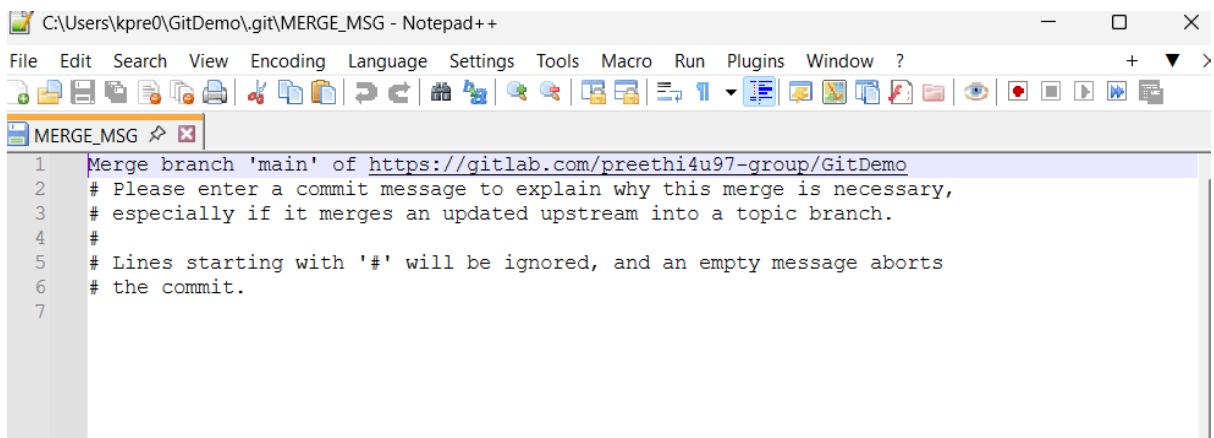
```
C:\Users\kpre0\GitDemo>git remote add origin https://gitlab.com/preethi4u97-group/GitDemo.git
```

```

C:\Users\kpre0\GitDemo>git pull origin main
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 2.74 KiB | 121.00 KiB/s, done.
From https://gitlab.com/preethi4u97-group/GitDemo
 * branch          main      -> FETCH_HEAD
 * [new branch]    main      -> origin/main
fatal: refusing to merge unrelated histories

C:\Users\kpre0\GitDemo>git pull origin main --allow-unrelated-histories
From https://gitlab.com/preethi4u97-group/GitDemo
 * branch          main      -> FETCH_HEAD
Merge made by the 'ort' strategy.
 README.md | 93 +++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
 1 file changed, 93 insertions(+)
 create mode 100644 README.md

```



```

PS C:\Users\kpre0\GitDemo> git -M
PS C:\Users\kpre0\GitDemo> git --allow-unrelated-histories
From https://gitlab.com/preethi4u97-group/GitDemo
 * branch          main      -> FETCH_HEAD
Already up to date.
PS C:\Users\kpre0\GitDemo> git -u
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 555 bytes | 277.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://gitlab.com/preethi4u97-group/GitDemo.git
 476bd2e..3cf4843  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\kpre0\GitDemo>

```

## Git HOL2

```
Windows PowerShell
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\kpre0\GitDemo> echo "This is a log file"
PS C:\Users\kpre0\GitDemo> mkdir

Directory: C:\Users\kpre0\GitDemo

Mode                LastWriteTime         Length Name
----                -
d-----           8/8/2025   1:50 PM         log

PS C:\Users\kpre0\GitDemo> echo "Inside log folder"
PS C:\Users\kpre0\GitDemo> notepad
PS C:\Users\kpre0\GitDemo> git
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\kpre0\GitDemo> git
PS C:\Users\kpre0\GitDemo> git -m "Add .gitignore to ignore .log files and log folder"
[main fa183e3] Add .gitignore to ignore .log files and log folder
 1 file changed, 2 insertions(+)
 create mode 100644 .gitignore
PS C:\Users\kpre0\GitDemo> git
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 348 bytes | 348.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://gitlab.com/preethi4u97-group/GitDemo.git
 3cf4843..fa183e3  main -> main
PS C:\Users\kpre0\GitDemo> |
```

### Git HOL3

```
Already on 'main'
Your branch is ahead of 'origin/main' by 4 commits.
  (use "git push" to publish your local commits)
PS C:\Users\kpre0\GitDemo> git branch GitNewBranch
fatal: a branch named 'GitNewBranch' already exists
PS C:\Users\kpre0\GitDemo> git branch -a
  GitNewBranch
* main
  remotes/origin/main
PS C:\Users\kpre0\GitDemo> git checkout GitNewBranch
Switched to branch 'GitNewBranch'
PS C:\Users\kpre0\GitDemo> echo "This file is from GitNewBranch" > branchfile.txt
PS C:\Users\kpre0\GitDemo> git add branchfile.txt
PS C:\Users\kpre0\GitDemo> git commit -m "Add branchfile.txt in GitNewBranch"
[GitNewBranch 8b4942e] Add branchfile.txt in GitNewBranch
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 branchfile.txt
PS C:\Users\kpre0\GitDemo> git status
On branch GitNewBranch
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml.orig

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\kpre0\GitDemo> git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 4 commits.
  (use "git push" to publish your local commits)
PS C:\Users\kpre0\GitDemo> git diff main GitNewBranch
diff --git a/.gitignore b/.gitignore
index 8c7c7f7..4a74afe 100644
Binary files a/.gitignore and b/.gitignore differ
diff --git a/branchfile.txt b/branchfile.txt
new file mode 100644
index 0000000..f7c9d4e
Binary files /dev/null and b/branchfile.txt differ
diff --git a/hello.xml b/hello.xml
deleted file mode 100644
```

```

3cf4843..fa183e3  main -> main
PS C:\Users\kpre0\GitDemo> git branch GitNewBranch
PS C:\Users\kpre0\GitDemo> git branch
  GitNewBranch
* main
PS C:\Users\kpre0\GitDemo> git checkout GitNewBranch
Switched to branch 'GitNewBranch'
PS C:\Users\kpre0\GitDemo> echo "This is content for the new branch" > newbranch.txt
PS C:\Users\kpre0\GitDemo> git add newbranch.txt
PS C:\Users\kpre0\GitDemo> git commit -m "Add file in GitNewBranch"
[GitNewBranch 8ba0958] Add file in GitNewBranch
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 newbranch.txt
PS C:\Users\kpre0\GitDemo> git status
On branch GitNewBranch
nothing to commit, working tree clean
PS C:\Users\kpre0\GitDemo> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\kpre0\GitDemo> git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\kpre0\GitDemo> git diff main GitNewBranch
diff --git a/newbranch.txt b/newbranch.txt
new file mode 100644
index 0000000..e5cd982
Binary files /dev/null and b/newbranch.txt differ

```

```

Binary files /dev/null and b/newbranch.txt differ
PS C:\Users\kpre0\GitDemo> git difftool main GitNewBranch

Viewing (1/4): '.gitignore'
Launch 'p4merge' [Y/n]? y

```

```

PS C:\Users\kpre0\GitDemo> git log --oneline --graph --decorate
* 545b45d (HEAD -> main) Ignore merge backup files
* 1e951ae Resolve merge conflict in hello.xml
|\
| * ab42ae2 Add hello.xml in GitWork
| * bd2615d Add hello.xml in main
|/
* fa183e3 (origin/main) Add .gitignore to ignore .log files and log folder
* 3cf4843 Merge branch 'main' of https://gitlab.com/preethi4u97-group/GitD
emo
|\
| * 476bd2e Initial commit
* 90d7fb5 Initial commit with welcome.txt
PS C:\Users\kpre0\GitDemo> |

```

```
PS C:\Users\kpre0\GitDemo> git branch -d GitNewBranch
error: the branch 'GitNewBranch' is not fully merged
hint: If you are sure you want to delete it, run 'git branch -D GitNewBranch'
hint: Disable this message with "git config advice.forceDeleteBranch false"
PS C:\Users\kpre0\GitDemo> |
```

```
PS C:\Users\kpre0\GitDemo> git status
On branch main
Your branch is ahead of 'origin/main' by 4 commits.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml.orig

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\kpre0\GitDemo> git push -u origin GitNewBranch
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 706 bytes | 706.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: To create a merge request for GitNewBranch, visit:
remote:   https://gitlab.com/preethi4u97-group/GitDemo/-/merge_requests/new?
merge_request%5Bsource_branch%5D=GitNewBranch
remote:
To https://gitlab.com/preethi4u97-group/GitDemo.git
 * [new branch]      GitNewBranch -> GitNewBranch
branch 'GitNewBranch' set up to track 'origin/GitNewBranch'.
```

```
nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\kpre0\GitDemo> git push -u origin GitNewBranch
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 706 bytes | 706.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: To create a merge request for GitNewBranch, visit:
remote:   https://gitlab.com/preethi4u97-group/GitDemo/-/merge_requests/new?
merge_request%5Bsource_branch%5D=GitNewBranch
remote:
To https://gitlab.com/preethi4u97-group/GitDemo.git
 * [new branch]      GitNewBranch -> GitNewBranch
branch 'GitNewBranch' set up to track 'origin/GitNewBranch'.
PS C:\Users\kpre0\GitDemo> git pull origin main
From https://gitlab.com/preethi4u97-group/GitDemo
 * branch            main       -> FETCH_HEAD
Already up to date.
```

## Git HOL4

```
gitNewBranch
* main
PS C:\Users\kpre0\GitDemo> git checkout GitNewBranch
Switched to branch 'GitNewBranch'
PS C:\Users\kpre0\GitDemo> echo "This is content for the new branch" > newbranch.txt
PS C:\Users\kpre0\GitDemo> git add newbranch.txt
PS C:\Users\kpre0\GitDemo> git commit -m "Add file in GitNewBranch"
[GitNewBranch 8ba0958] Add file in GitNewBranch
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 newbranch.txt
PS C:\Users\kpre0\GitDemo> git status
On branch GitNewBranch
nothing to commit, working tree clean
PS C:\Users\kpre0\GitDemo> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\kpre0\GitDemo> git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\kpre0\GitDemo> git diff main GitNewBranch
diff --git a/newbranch.txt b/newbranch.txt
new file mode 100644
index 0000000..e5cd982
Binary files /dev/null and b/newbranch.txt differ
PS C:\Users\kpre0\GitDemo> git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\kpre0\GitDemo> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\kpre0\GitDemo> git branch GitWork
PS C:\Users\kpre0\GitDemo> git checkout GitWork
Switched to branch 'GitWork'
PS C:\Users\kpre0\GitDemo> echo "<greeting>Hello from GitWork branch</greeting>" > hello.xml
PS C:\Users\kpre0\GitDemo> git add hello.xml
PS C:\Users\kpre0\GitDemo> git commit -m "Add hello.xml in GitWork"
```

```

PS C:\Users\kpre0\GitDemo> git commit -m "Add hello.xml in GitWork"
[GitWork ab42ae2] Add hello.xml in GitWork
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 hello.xml
PS C:\Users\kpre0\GitDemo> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\kpre0\GitDemo> echo "<greeting>Hello from MAIN branch</greeting>" > hello.xml
PS C:\Users\kpre0\GitDemo> git add hello.xml
PS C:\Users\kpre0\GitDemo> git commit -m "Add hello.xml in main"
[main bd2615d] Add hello.xml in main
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 hello.xml
PS C:\Users\kpre0\GitDemo> git config --global merge.tool p4merge
PS C:\Users\kpre0\GitDemo> git config --global mergetool.p4merge.path "C:/Program Files/Perforce/p4merge.exe"
PS C:\Users\kpre0\GitDemo> git config --global diff.tool p4merge
PS C:\Users\kpre0\GitDemo> git config --global difftool.p4merge.path "C:/Program Files/Perforce/p4merge.exe"
PS C:\Users\kpre0\GitDemo> git log --oneline --graph --decorate --all
* bd2615d (HEAD -> main) Add hello.xml in main
| * ab42ae2 (GitWork) Add hello.xml in GitWork
|/
| * 8ba0958 (GitNewBranch) Add file in GitNewBranch
|/
* fa183e3 (origin/main) Add .gitignore to ignore .log files and log folder
* 3cf4843 Merge branch 'main' of https://gitlab.com/preethi4u97-group/GitDemo
| \
| * 476bd2e Initial commit
| * 90d7fb5 Initial commit with welcome.txt
PS C:\Users\kpre0\GitDemo> git diff main GitWork
diff --git a/hello.xml b/hello.xml
index 855d11d..d423c28 100644
Binary files a/hello.xml and b/hello.xml differ
PS C:\Users\kpre0\GitDemo> git difftool main GitWork

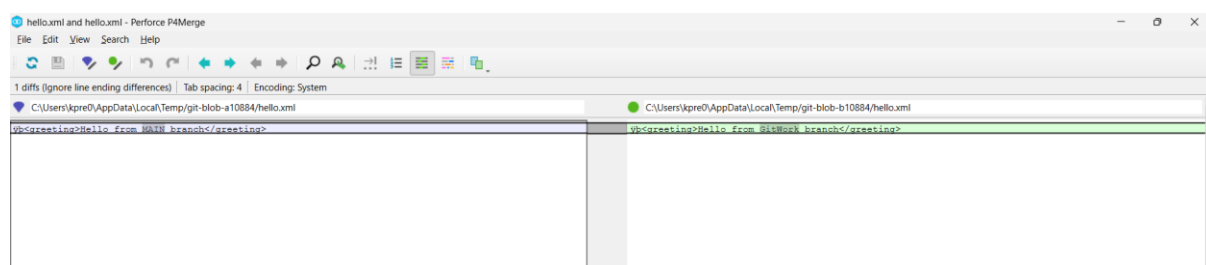
Viewing (1/1): 'hello.xml'
Launch 'p4merge' [Y/n]? y

```

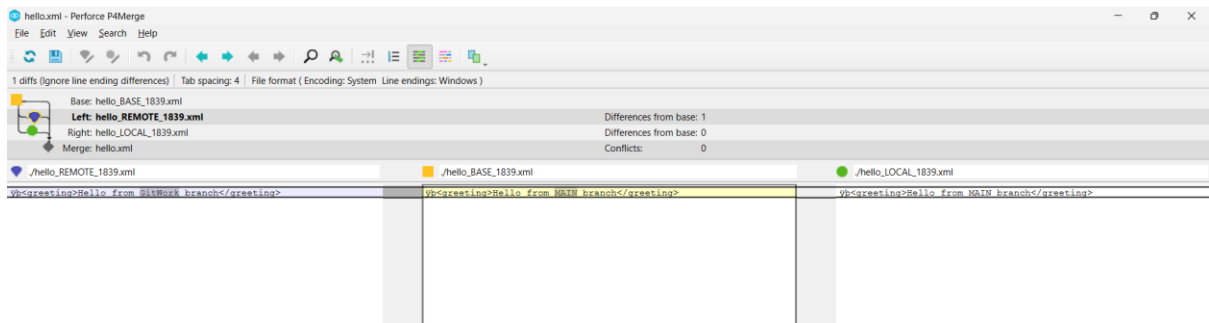
```

PS C:\Users\kpre0\GitDemo> git merge GitWork
warning: Cannot merge binary files: hello.xml (HEAD vs. GitWork)
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

```







```
PS C:\Users\kpre0\GitDemo> git mergetool
Merging:
hello.xml

Normal merge conflict for 'hello.xml':
  {local}: created file
  {remote}: created file
error: No valid patches in input (allow with "--allow-empty")
PS C:\Users\kpre0\GitDemo> git add hello.xml
PS C:\Users\kpre0\GitDemo> git commit -m "Resolve merge conflict in hello.xml"
[main 1e951ae] Resolve merge conflict in hello.xml
PS C:\Users\kpre0\GitDemo> echo "*.orig" >> .gitignore
PS C:\Users\kpre0\GitDemo> git add .gitignore
PS C:\Users\kpre0\GitDemo> git commit -m "Ignore merge backup files"
[main 545b45d] Ignore merge backup files
 1 file changed, 0 insertions(+), 0 deletions(-)
PS C:\Users\kpre0\GitDemo> git branch -d GitWork
Deleted branch GitWork (was ab42ae2).
PS C:\Users\kpre0\GitDemo> git log --oneline --graph --decorate
* 545b45d (HEAD -> main) Ignore merge backup files
* 1e951ae Resolve merge conflict in hello.xml
|\
| * ab42ae2 Add hello.xml in GitWork
* | bd2615d Add hello.xml in main
|/
* fa183e3 (origin/main) Add .gitignore to ignore .log files and log folder
* 3cf4843 Merge branch 'main' of https://gitlab.com/preethi4u97-group/GitDemo
PS C:\Users\kpre0\GitDemo>
```

## Git HOL5

```
PS C:\Users\kpre0\GitDemo> git branch -a
GitNewBranch
* main
  remotes/origin/GitNewBranch
  remotes/origin/main
PS C:\Users\kpre0\GitDemo> git pull origin main
From https://gitlab.com/preethi4u97-group/GitDemo
 * branch                main          -> FETCH_HEAD
Already up to date.
```

```
PS C:\Users\kpre0\GitDemo> git pull origin main
From https://gitlab.com/preethi4u97-group/GitDemo
 * branch                main          -> FETCH_HEAD
Already up to date.
PS C:\Users\kpre0\GitDemo> git push origin main
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 8 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (12/12), 1.23 KiB | 632.00 KiB/s, done.
Total 12 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
To https://gitlab.com/preethi4u97-group/GitDemo.git
 fa183e3..545b45d  main -> main
```